PROJECT REPORT-THE KOZHIKODE BY-PASS PROJECT

IFC'S PERFORMANCE STANDARD 7- INDIGENOUS PEOPLE

COURSE NAME:

ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENT

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1. International Finance Corporation's Performance Standard 7

Aiming at indigenous people and the effects development projects have on them ***IFC Performance Standard 7 (PS 7)*** suggests possible mitigation measures to curb such effects. Further, it is directed towards the protection of their rights, protection of their cultural heritage and ownership rights. It promotes equitable and sustainable benefits from projects affecting their lands, resources, and communities. Such groups are dependent on their natural resources, when resettled they lose their identity and become vulnerable.

1.1. IDENTIFYING AND DEFINING INDIGENOUS PEOPLES

The first step would require identifying indigenous groups. There is no universally accepted definition of indigenous people. Along with People identified by national law as indigenous it will also include social groups self-identified and recognized by others as distinct, with a collective attachment to geographically specific habitats, unique cultural practices, and often separate languages or dialects. It even includes nomadic tribes and people who have been displaced but have continued cultural and spiritual ties with their ancestral land.

Indigenous people and vulnerability to Land and Resource Loss: Indigenous Peoples are highly vulnerable when their lands are infringed upon, as their livelihoods, identities, and traditional knowledge systems are inextricably linked to these territories and natural resources. Their cultures and oral histories are maintained by links with sacred and significant sites.

When these are damaged or modified due to any development project in a way that they can't use or maintain their traditional knowledge, economic stability, and cultural heritage, this loss of livelihood and cultural identity makes them vulnerable as, unlike mainstream people, they can't maintain their economic stability.

1.2. OBJECTIVES OF PS 7

- **1.2.1.** Ensuring full respect for Indigenous Peoples' human rights, dignity, culture, and livelihoods.
- **1.2.2.** Avoiding or minimizing adverse impacts from development projects, with compensation when impacts are unavoidable.
- **1.2.3.** Promoting culturally appropriate, sustainable development benefits.
- **1.2.4.** Establishing an ongoing relationship with Indigenous communities through **Informed Consultation and Participation (ICP)**.
- **1.2.5.** Securing **Free, Prior, and Informed Consent (FPIC)** under specified circumstances.

1.3. SCOPE OF APPLICATION

PS 7 applies when a project affects Indigenous communities, particularly when land or resources are crucial to their identity or livelihoods. If a land resource falls under national jurisdiction and confers land rights or customary rights to tribes, this PS will apply. It will apply even if indigenous people do not dwell in the area and just consume the resources for generations.

The magnitude of the effect is assessed by environmental and social risk assessments, which are guided by expert analysis. It is also applicable if cultural heritage such as sacred groves, sacred bodies of water and waterways, sacred trees, and sacred rocks are involved; cultural heritage is further guided in **PS 8**.

1.4. ENGAGEMENT AND CONSENT

Informed Consultation and Participation (ICP): Indigenous communities must be actively engaged in decision-making through a transparent and respectful process. It needs to be done as required in ***PS 1***.

1.5. PARTICIPATION, CONSENT, AND FPIC

Free, Prior, and Informed Consent (FPIC) includes consultation and requires good-faith negotiation and agreement. ***FPIC*** is mandatory in cases involving impacts on traditional or customary lands and resources, relocation from traditional territories, and significant effects on critical cultural heritage.

The process ensures that Indigenous communities understand project risks and benefits and are empowered to make decisions.

1.6. AVOIDANCE AND MITIGATION OF IMPACT

1.6.1. *Avoidance*:

Development Projects must find alternatives to avoid harming Indigenous Peoples' lands, resources, or cultural identity.

1.6.2. *Mitigation and Compensation*:

Where alternatives can't be found, culturally appropriate measures need to be taken to restore livelihoods or compensate communities. These measures should align with the vulnerability and needs of the affected groups. The compensation could be either land-based or cash compensation and identify alternative livelihood. Land-based compensation is preferred, and only if it is not feasible should monetary compensation be provided.

While relocating them, ensuring continued natural resources, and identifying equivalent replacements should be done. Compensation can be individual or collective or both, reflecting the community's customs and governance. Often, women are more involved and manage natural resources; their vulnerability should not be neglected. When compensation is on a collective basis, then enough measures should be taken to ensure effective distribution.

The clients involved in the development project must actively engage with Indigenous Peoples and government authorities, addressing gaps between national laws and IFC standards where feasible

2. Detailed Project Description of the Kozhikode Bypass Project

2.1. PROJECT AND PUBLIC GOALS

The Kozhikode Bypass project which is originally **part of the National Highway 66 (NH 66)** project connecting Kanyakumari to Panvel, the infrastructural project is essentially designed to tackle the persistent issues of ***traffic congestion*** and safety in Kozhikode.

The main objective of this project is to ***upgrade the current two-lane highway*** into ***a six-lane bypass corridor***, offering commuters a safer and more efficient transportation route. This would reduce the existing traffic difficulties in the heavily populated regions of ***Malaparamba***, ***Thondayad***, and ***Ramanattukara***, which have historically suffered from congestion and accidents.

The project facilitates better traffic movement as well as improves the connectivity and accessibility of the area, benefiting both local inhabitants and long-distance travellers by reducing their time taken for commutes utilizing the NH 66 corridor.

2.2. LOCATION

The Bypass project **stretches from Vengalam to Ramanattukara** in Kozhikode district, covering approximately 28.4 kilometres, making it a substantial infrastructure project with important implications for the regional transportation network.

2.3. LAND REQUIRED FOR THE PROJECT

The expansion of the bypass into six lane highway project required total **land area of 127.80 hectares**, with **125.94 hectares of this land being available NH land**. This land allocation will enable the construction of the expanded six-lane highway, including the main carriageway, service roads on both sides, and the necessary supporting infrastructure such as flyovers, underpasses, culverts, and other ancillary facilities.

2.4. PROJECT CONSTRUCTION PROGRESS

The construction work for the Kozhikode Bypass project commenced by mid-April 2018, the contract was given to Hyderabad-based Krishna Mohan Construction Private Limited (KMC) for a total project cost of ₹1,424 crore accounting 42.79 crore per km. The contractor had to submit a bank guarantee of ₹85.50 crore and sign the agreement within 45 days, paving the way for the start of construction activities.

The comprehensive project report for the bypass has been granted technical approval, signifying that all required planning and permissions have been secured, thereby accelerating the transition to the implementation phase. Nearly 70% of the project is complete and it is expected to be open for the public by January 2025.

2.5. WORKFORCE REQUIREMENT

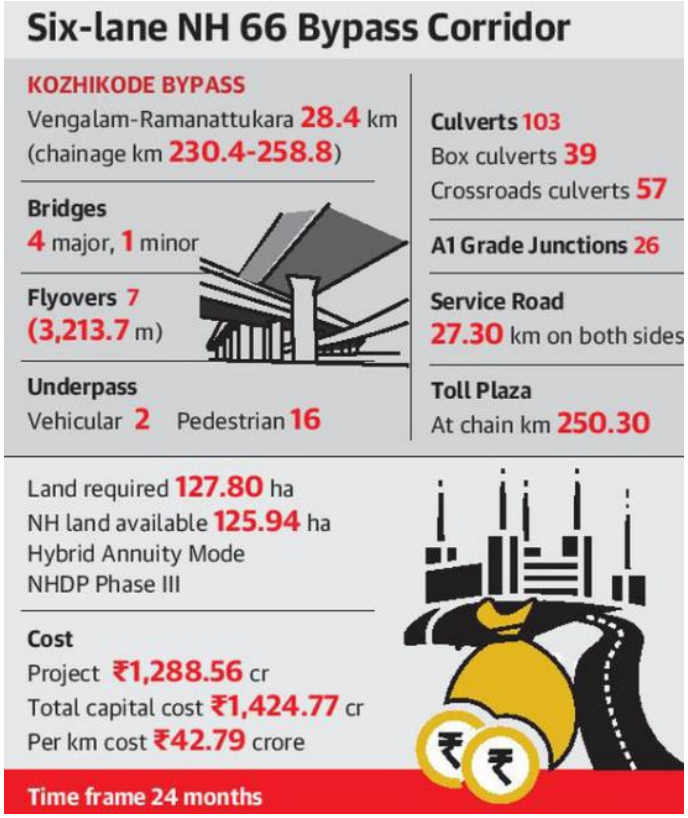
For the successful execution of the Bypass project required a substantial workforce, comprising engineers, technicians, skilled and unskilled labourers, and support staff. The exact number of workers needed was determined based on the construction schedule and the various stages of the project.

The contractor, ***Krishna Mohan Construction Private Limited***, had the responsibility for sourcing and managing the required workforce to ensure the timely and efficient completion of the bypass expansion. This influx of employment opportunities will result in a positive impact on the local economy, generating jobs and supporting the livelihoods of the surrounding communities.

2.6. NEED FOR ANCILLARY INFRASTRUCTURAL FACILITIES

In addition to the main six-lane highway, the Kozhikode Bypass project will include the development of other supporting infrastructural facilities. These includes 4 major bridges, 1 minor bridge, 7 flyovers, 2 vehicle underpasses, 16 pedestrian underpasses, 64 pipe culverts, and 39 box culverts.

These ancillary structures will play crucial role in strengthening the bypass's overall functionality and connectivity thereby boosting traffic flow and assuring car and pedestrian safety. The integration of these supporting elements will build a comprehensive transportation network that can effectively handle the present issues encountered by commuters and local inhabitants in the Kozhikode area.



3. Social Receptors and Interaction with the Project

In the context of the Kozhikode bypass project in Kerala, social receptors refer to individuals or groups that may be affected by the project's implementation. Given that there is no distinct Indigenous community directly impacted by the bypass, the focus shifts to nearby populations, including fringe communities, who may experience various social interactions because of this infrastructure development.

3.1. KEY SOCIAL RECEPTORS

3.1.1. Fringe Communities

These include villagers living in proximity to the bypass route, such as those in Ellatur. They may experience changes in their daily lives due to increased traffic, noise pollution, and alterations in local land use.

3.1.2. Fishing Communities

Local fishing populations may be affected by changes in water quality or access to fishing grounds resulting from construction activities. The increased traffic and potential pollution could disrupt their traditional livelihoods.

3.1.3. Nomadic Tribes

Nomadic groups that traverse the region may face disruptions in their traditional routes and access to resources due to the construction and subsequent operation of the bypass.

3.1.4. Small Shop Owners

Local businesses situated along or near the bypass may experience changes in customer flow due to altered traffic patterns. While some businesses might benefit from increased accessibility, others could suffer from reduced foot traffic during construction phases.

3.2. NATURE OF INTERACTIONS

The interactions between these social receptors and the project can lead to both positive and negative outcomes. Understanding these dynamics is crucial for effective project planning and implementation.

3.2.1. Positive Interactions

i. **Economic Opportunities**: Improved transportation infrastructure can lead to enhanced access to markets for local businesses, potentially increasing sales and economic activity in the area.

ii. **Job Creation**: During the construction phase, local labor may be employed, providing temporary job opportunities for residents of fringe communities.

iii. **Improved Connectivity**: The bypass is expected to reduce travel time for commuters, benefiting residents who rely on efficient transportation for work and daily activities.

3.2.2. Negative Interactions

i. **Disruption of Daily Life**: Increased traffic during construction can lead to safety concerns for residents and disrupt their daily routines. Noise and dust from construction activities may also affect their quality of life.

ii. **Cultural Displacement Risks**: Changes in land use could threaten local cultural practices or heritage sites associated with nearby communities. This necessitates careful monitoring and protection efforts.

iii. **Social Tensions**: If concerns regarding land acquisition or environmental impacts are not adequately addressed through engagement initiatives, tensions may arise between project developers and local communities.

3.3. ENGAGEMENT STRATEGIES

To mitigate potential negative impacts and enhance positive interactions, it is essential to implement effective community engagement strategies:

i. **Regular Communication**: Establishing open lines of communication with affected communities will help address concerns promptly and foster trust between stakeholders.

ii. **Consultation Processes**: Involving local residents in consultation processes will ensure their voices are heard in decision-making related to the project.

iii. **Monitoring Programs**: Implementing monitoring frameworks will allow for ongoing assessment of social impacts throughout all phases of the project, ensuring that any emerging issues are addressed proactively.

4. Baseline of the Project

Baseline report is important for assessing the potential impact of the six laning projection the social, environmental and economic aspects. This report is the foundation for further assessment including Social Impact Assessment.

4.1. SOCIAL AREA OF INFLUENCE (SAOI)

SAOI is the area where the project can impact directly or indirectly which includes the local communities along the route.

4.2. DEMOGRAPHIC PROFILE

For evaluating the social impact of the construction, it is important to understand the demography of the population.

4.2.1. Population Characteristics

i. **Total Population**: Approximately 783 residents. (**Source:** Vengalam Municipality**)**

ii. **Age Distribution**:

a. Children (0-14 years): 30%

b. Working Age (15-64 years): 65%

c. Elderly (65 years and above): 5%

iii. **Sex Ratio** : 1.097: 1 (Female: Male)

iv. **Religion**

Religion	Percentage (%)
Hindu	55.67%
Muslim	42.34%
Christian	1.99%

***Source**: Secondary data acquired from Ramanattukara Municipality (Janakeeya Asoothranam Report 2023)*

4.2.2. Social Category

Religion	Percentage (%)
GEN	49.36%
OBC	48.25%
SC/ST	2.39%

4.2.3. Socio-Economic Status

i. **Income Levels**:

There are differences in income between the local people. A portion of the population lives below the poverty line and depends on low-paying occupations or informal labor, even while some families have reasonably stable employment.

ii. **Employment**:

OCCUPATION	PERCENTAGE (%)
Small scale business & Street vendors	24.33
Agriculture	17.00
Fisheries	12.18
Commuters	13.53
Private jobs	32.96

4.3. DIRECTLY AFFECTED COMMUNITIES

Primary Stakeholders include residents and businesses alongside the route. Construction activities and increment of noise and pollution levels. Due to limited accessibility during the construction phase, the bypass may also cause changes in local business, a decline in community relations, and variations in property values.

4.3.1. Villagers and Fringe Communities

Villagers primarily rely on **agriculture, small trades, and livestock** for their livelihood. The baseline assessment includes demographic details, economic activities, and access to essential services.

Indicator	Description	Estimated Figures
Population	Total village population	3,000 people
Household Size	Average number of individuals per household	5.2
Employment Rate	Percentage of adults employed	65% (agriculture: 40%, small trades: 25%)
Education Levels	Primary, Secondary, Tertiary	Primary: 70%, Secondary: 20%, Tertiary: 10%
Average Income	Monthly income per household	20000 rs

Access to Healthcare	Percentage of population with access to basic healthcare facilities	60%
Land Ownership	Average land owned per household	1.5 acres
Resource Dependency	Primary resources used in agriculture (water, grazing land)	High dependency on local water sources
Cultural Sites	Number of sites used for community events and gatherings	3 community centers

INFERENCE: *Villagers are moderately dependent on land for agriculture, which provides food and income. Access to healthcare and education is limited, and the community holds periodic gatherings at specific cultural sites. Potential disruptions to land access or water resources could significantly impact their livelihoods and social practices.*

4.3.2. Small Shop Owners

Small shop owners in the region serve the local population, providing goods and services. Their baseline data includes economic indicators, customer base reliance, and supply chain dependencies.

Indicator	**Description**	**Estimated Figures**
Population	Total population engaged in small shop ownership	200 people
Average Monthly Income	Monthly revenue per shop	$300 USD
Customer Base	Percentage of customers from local villagers and communities	70%
Primary Goods Sold	Common items stocked (groceries, tools, household items)	Groceries: 60%, Household: 30%, Misc: 10%
Supply Chain Dependency	Number of suppliers and average distance to suppliers	3 main suppliers, 1-hour distance
Impact of Construction	Expected revenue drop during construction phase	Estimated 25% drop
Shop Ownership	Percentage of locally owned shops	90%
Credit Dependency	Percentage of shops offering customer credit	40%
Family Dependency	Percentage of family members working in shops	50%
Access to Banking	Percentage with access to financial services	30%

***INFERENCES**: Small shop owners are highly dependent on the local population, with about 70% of their customer base drawn from surrounding communities. Revenue fluctuations can significantly affect family income, as many shops are family-run businesses. Supply chain delays during construction could affect inventory, impacting sales and reducing revenue. Access to financial services is limited, meaning shops rely heavily on local transactions and lack financial resilience to withstand prolonged disruptions.*

4.4. INDIRECTLY AFFECTED COMMUNITIES

The construction of the six lanes will affect the neighbourhoods where change in traffic patterns will lead to change in economic activities.

4.4.1. Fishing Communities

The fishing community is closely tied to **local water bodies**, which serve as a source of both sustenance and income. The following baseline data highlights the key socioeconomic and environmental factors relevant to their livelihood.

Indicator	Description	Estimated Figures
Population	Total population engaged in fishing activities	1,200 people
Household Size	Average number of individuals per household	4.5
Primary Income Source	Percentage of population depending solely on fishing	80%
Average Monthly Income	Monthly income per household	20000 rs
Education Levels	Primary, Secondary	Primary-60%, Secondary-15%
Water Resource Dependency	Proportion of income directly reliant on water access	90%
Boat Ownership	Percentage of households owning fishing boats	45%
Fish Stock Levels	Estimated annual catch in kg (per household)	800 kg per household
Market Access	Distance to nearest market (average travel time)	45 minutes
Cultural Significance	Number of fishing-related ceremonies held annually	2 major ceremonies

***INFERENCES**: Fishing communities have a high dependency on local water bodies for their income and food security. Any restrictions on water access or resource depletion could drastically impact their financial stability. Cultural ties to fishing are evident in traditional ceremonies linked to the seasonal fishing cycle, emphasizing the need to maintain access to fishing areas.*

4.4.2. Nomadic Tribes

Nomadic tribes in the region rely on **seasonal migration patterns** for grazing and trade, with cultural practices tied to specific lands and routes. The baseline for these groups includes data on migration routes, resource dependencies, and social structure.

Indicator	Description	Estimated Figures
Population	Estimated nomadic tribe population	800 people
Household Size	Average number of individuals per household	6.0
Migration Frequency	Number of annual migrations per tribe	3 seasonal movements
Grazing Dependency	Percentage of livestock-dependent families	85%
Average Herd Size	Number of animals per household	30 animals (mixed livestock)
Cultural Sites	Number of sacred or culturally significant sites	5 sites
Education Access	Percentage of children with access to schooling	20%
Primary Income Source	Percentage of population dependent on livestock and craft sales	70%
Access to Health Facilities	Proportion with regular access to healthcare	15%
Water Source Access	Number of wells or water sources per migration route	2 main water sources

***INFERENCES**: Nomadic tribes follow traditional migration routes essential for grazing and maintaining their cultural identity. Access to water sources and grazing land is crucial for their livestock, which serves as their primary income. Restricted access to these routes would threaten their lifestyle, cultural heritage, and financial well-being. Additionally, their low access to healthcare and education underscores their vulnerability.*

4.5. ECONOMIC CONDITIONS

Since much of the population works in small-scale company, service-oriented occupations, and agriculture, the area where the project will be executed has a diverse economy.

The main crops that support many households include coconut, cocoa, nutmeg, black and white pepper, allspice, ginger, cashew nuts, rubber, bananas, and sugar cane. Potentially detrimental elements such frequent dust, contaminated water, or changed access points could affect agricultural output during the building phase, endangering jobs in the agricultural sector.

Like this, depending on changes in traffic patterns and rerouting, small businesses that are heavily frequented by the locals may experience temporary stagnation, losses, or new growth chances. Such shifts clearly show the necessity of careful protections against economic shocks to avoid losses and to create new beneficial sources of development in the communities.

4.6. PROPERTY VALUES

Because major highways are more accessible after construction, property values in these locations typically rise. Nonetheless, brief decreases are typical during construction, particularly in regions impacted by dust, noise, and traffic jams.

Area	**Property Value Impact**
Adjacent to Bypass	Likely Increase Post-Construction
Near Construction Zones	Temporary Decline

5. Potential Impact of the Project Across the Life Cycle

The project's impact evolves over the life cycle of the project, starting from planning phase till the operational phase. Each phase has a set of challenges that interact with the livelihood, cultural practices and well-being of the affected communities.

5.1. PLANNING PHASE

The impacts during the planning phase are mainly dependent on the adequacy of engagement of stakeholders and assessment of resources. Poor planning can lead to mistrust, exclusion of vulnerable groups, and improper preparation for mitigation.

5.1.1. Stake holder Exclusion

The failure of involving villagers, fishing communities, fringe communities and nomadic tribes in early consultations can lead to lack of community buy in and project resistance.

5.1.2. Inadequate resource mapping

Neglecting the importance of resources like fishing grounds, grazing area / agricultural lands can lead to poor management in future which would have immediate impact on people that depend on them.

5.1.3. Grievance Redress

Lack of proper methods for addressing concerns may lead early grievance to increase, delaying project implementation.

5.2. CONSTRUCTION PHASE

Often the most destructive phase due to physical activities due to land clearance, water diversion, construction activities, increased vehicular movements etc.

5.2.1. Displacement and loss of land

Villagers will lose access to their agricultural land impacting their livelihood directly. Some of the shopkeepers near by the road will lose their shop due to demolitions, others may experience reduced consumer traffic.

5.2.2. Restricted Access to Resources

The fishing communities may lose access to their fishing grounds, while nomadic tribes might lose their migration routes and cultural space.

5.2.3. Environmental Destructions

The noise, dust, and pollution will affect the health and wellbeing of the nearby communities, and agricultural crops.

5.2.4. Cultural site Disruption

The construction activities may damage or block the cultural and religious places, thereby interrupting festivals and traditional practices.

5.2.5. Traffic & Safety Issues

Increased vehicular movements can disturb the local traffic, and can cause safety risk for pedestrians, livestocks, animals and nomadic tribes.

5.3. OPERATIONAL PHASE

In this phase the impacts become more systemic and long-term affecting the socioeconomic pattern.

5.3.1. Permanent changes in land & resources

The changes in water bodies and land use might affect fishing activities, agriculture and migration patterns permanently.

5.3.2. Economic Shifts

The new infrastructure may increase the accessibility and opportunities like market access. But at the same time, it may displace the traditional livelihood or increase the competition from larger businesses.

5.3.3. Cultural Transformations

The communities may lose touch of their culture amidst the rise of development and instead transcend into modern mixes of their own culture and the world outside.

5.3.4. Environment Degradation

May decrease the water quality, air quality index, increase the noise pollution and degrade the local ecosystem, that further affects the livelihood of people.

Each of the project phases has its own mode of risks and opportunities, the construction phase with most immediate destruction and the operational phase with more systemic and long-lasting effects. Sustained and early mitigation measures combined with strong community engagement is essential to minimize the negative impacts and maximizing positive outcomes throughout the life cycle of the project.

6. Impact identification and Significance

The following table summarizes the identified impacts based on their significance, stage of impact, characteristics, and overall significance rating. This table helps to provide a clear overview of the potential effects of a project on various social groups.

Sl No:	Identified Impact	Stage of Impact	Impact Characteristics	Impact Significance
1.	Agricultural Land Loss	Construction	Agriculture is the primary livelihood for many villagers, permanent loss of land due to land acquisition. Affecting specifically local farming communities and loss of agricultural assets like coconut trees and other plantation crops adjacent to the project area.	Moderate to High
2.	Bifurcation of village	Construction	Due to the construction of the bypass certain villages are divided into two halves pedestrians face difficulties in crossing the bypass and reach other side due to lack of underpass. Mainly due to delay in construction activity. Locals are affected increasing their travel expense.	Moderate to High

3.	Disruption of Routes	Pre-Construction	If routes are permanently restricted or altered due to delay in construction activity, Regional impact, affecting multiple groups relying on similar paths.	High
4.	Loss of Cultural and Ancestral Sites	Pre-Construction	Sites hold historical and cultural importance; destruction impacts identity and heritage. Cultural institutions like mosque and temples and a sacred grooves having religious sentiments affected. Regional impact which can be Permanent if sites are destroyed.	High
5.	Economic Displacement	Construction	Shop owners rely on customer traffic, which could reduce during construction, lowering income levels. Also the small scale vendors have to be relocated due to land acquisition. During construction but could extend Long-Term if customer access remains impeded.	Moderate to High

6.	Isolation of village	Post construction	Before the bypass the accessible road was NH 17 passing through - Elathur village a fishing village with tourism based activity for livelihood. New bypass reduce the travellers count choosing the old NH 17 road tourism as well as livelihood of people will be affected	High
7.	Increased Competition and Market Changes	Post-Construction	New businesses may increase competition for small shop owners since they have to start a new business elsewhere. - Long-Term, persisting issue as long as new businesses operate. -	Moderate
8.	Water Logging and Drainage Issues	Construction	Disruption in local drainage patterns may lead to waterlogging, particularly during heavy rains.	**Moderate** – Direct, Short-term
9.	Improved Connectivity and Economic Opportunities	Operation	Increased access to markets, services, and employment opportunities; Rise in property values.	**High** – Indirect, Long-term
10.	Planned Urban Development and Increased Land Value	Operation	Enhanced opportunities for urban planning and investment in local infrastructure and services.	**High** – Indirect, Long-term

11.	Risk of Increased Traffic Accidents	Operation	Higher traffic volume leading to safety concerns, particularly for pedestrians and vulnerable road users.	**Moderate** – Direct, Long-term
12.	Increased Vulnerability for Women and Children	Construction/Operation	Potential rise in exploitation or insecurity, particularly during construction activities or due to new inflows of population.	**High** – Direct/Indirect, Medium to Long-term
13	Resource Depletion (Fish Stock)	Construction	Fishing is central to certain communities nearby the highway livelihoods; depletion threatens food security and income. - **Duration**: Long-Term, possibly extending through the project lifecycle. - **Extent**: Local to Regional, affecting other nearby fishing communities. - **Reversibility**: Irreversible	High

7. Mitigation measures and expected outcomes

7.1. VILLAGERS AND LOCAL COMMUNITIES

- **Mitigation Measures**

i. **Cultural site preservation:** With informed consultation Identify cultural site, which is at risk, provide temporary routes during construction.

ii. **Community Participation:** Prepared a structured participation process with various local representatives.

iii. **Compensation for Crop loss:** Fair Financial compensation for damaged crops, which is based on current market rate of crop.

iv. **Alternative Land:** Restore equal size with of plots, with good soils fertility.

- **Expected Outcomes**

i. Less social disturbance, safeguarding identity of peoples
ii. Meaningful discussions, to strengthen fairness.
iii. Improvement in crop productivity.

7.2. INDIGENOUS TRIBES

- **Mitigation Measures**

i. **Heritage site protection**: Identify cultural or religious sites and provides safeguarding recommendation.

ii. **Community Consultation**: Always consult with tribes' representative to reduce conflict.

iii. **Alternative routes**: Analysing & developing possible alternative routes to make access continued.

- **Expected Outcomes**

i. Reserving culture, heritage site.
ii. Ensure their role in decision making.
iii. Reduce difficulties.

7.3. FISHING COMMUNITIES

- **Mitigation Measures**

i. **Water quality**: Setting up regular water quality monitoring Station. During construction & operation, apply possible measures to prevent pollution.
ii. **Fishing Access**: Provide temporary fishing access points before any project activity begins.
iii. **Restocking & conservation measures**: Allying with fisheries to restock affected area win fish.
iv. **Compensation**: Monetary compensation equal to income lost during the fishing period.

- **Expected Outcomes**

i. Improved aquatic ecosystem.
ii. Maintained accessible road.
iii. Financial assistance.
iv. Strengthening of community resilience.

7.4. SHOP KEEPERS

- **Mitigation Measures**

i. **Temporary routes**: Construction of temporary pathways to ensure customer access.
ii. **Skill Development**: Arrange training sessions on marketing strategies, financial management, etc.
iii. **Assistance**: Provide grants or low-interest loans.

- **Expected Outcomes**

i. Bringing more customer towards small shopes.
ii. Handling of situations in changing market effectively.
iii. Strengthen local economy.

7.5. DRAINAGE & WATERLOGGING

- **Mitigation Measures**

i. **Drainage solutions**: Draw layout & construct Improved drainage system.
ii. **Drainage assessment**: Analyse & study of drainage across a project site.
iii. **Regular Monitoring**: Regular analysis of drainage system.
iv. **People's involvement**: Engage with locals to ensure reporting of any issue during construction.

- **Expected Outcomes**

i. Prevention from water logging.
ii. Reduced soil erosion.
iii. Less harm to local environment.

7.6. TRAFFIC ACCIDENTS

- **Mitigation Measures**

i. **Traffic Management**: Installation of sign board, speed breaker to enhance safety.
ii. **Awareness Campaigns**: Emphasis on road safety, encourage peoples to wear seat belt & helmet.
iii. **Audits**: Recognise high-risk areas and strengthening management.

- **Expected Outcomes**

i. Reduction in accidents.
ii. People are more aware about their safety.
iii. Improved collaboration among locals & project developers.

7.7.Cross cutting

- **Mitigation Measures**

i. **Stakeholder Participation**: Regular public Meeting, Structured framework.
ii. **Monitoring**: Effective monitoring system to improve mitigation measures.

- **Expected Outcomes**

i. Build trust through communication.
ii. Proactive engagement with various stakeholders.

8. References

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